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I. Translate the given sentences into logical expressions.

1.Let p, q , and r be the propositions

p : Grizzly bears have been seen in the area.

q : Hiking is safe on the trail.

r : Berries are ripe along the trail.

a. Berries are ripe along the trail, but grizzly bears have not been seen in the area.

$r \land \neg p$

b. If berries are ripe along the trail, hiking is safe if and only if grizzly bears have not been seen in the area.

$r \rightarrow (q \leftrightarrow \neg p)$

c. Hiking is not safe on the trail whenever grizzly bears have been seen in the area and berries are ripe along the trail.

$(p \land r) \rightarrow \neg q$

2.Let p, q, and r be the propositions

p : You get an A on the final exam.

q : You do every activity in this class.

r : You get an A in this class.

a. You will get an A in this class unless you do not get an A on the final exam.

$r \lor \neg p$ (equivalent: $\neg p \rightarrow r$)

b. You will get an A in this class if and only if you do every activity or you get an A on the final exam.

$r \leftrightarrow (q \lor p)$

II. Construct the truth tables for each compound proposition.

1. $\mathbf{p \lor q \to \neg p \land \neg q}$
2. $\mathbf{(p \to q) \oplus (\neg p \leftrightarrow q) \land \neg q}$
3. $\mathbf{(p \leftrightarrow \neg q) \to \neg (q \lor r)}$

p	q	r	$p \lor q$	$\neg p$	$\neg q$	$\neg p \land \neg q$	$p \lor q \to \neg p \land \neg q$	$p \to q$	$\neg p \leftrightarrow q$	$(p \to q) \oplus (\neg p \leftrightarrow q)$	$(p \to q) \oplus (\neg p \leftrightarrow q) \land \neg q$	$p \leftrightarrow \neg q$	$q \lor r$	$\neg(q \lor r)$	$(p \leftrightarrow \neg q) \to \neg(q \lor r)$
T	T	T	T	F	F	F	F	T	F	T	F	F	T	F	T
T	T	F	T	F	F	F	F	T	F	T	F	F	T	F	T
T	F	T	T	F	T	F	F	F	T	T	T	T	T	F	F
T	F	F	T	F	T	F	F	F	T	T	T	T	F	T	T
F	T	T	T	T	F	F	F	T	T	F	F	T	T	F	F
F	T	F	T	T	F	F	F	T	T	F	F	T	T	F	F
F	F	T	F	T	T	T	T	T	F	T	T	F	T	F	T
F	F	F	F	T	T	T	T	T	F	T	T	F	F	T	T